SANKUR KUNDU

Curriculum Vitae | ML Systems Engineer & Independent Researcher
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SUMMARY

ML systems engineer and independent researcher (B.Tech CSE, 2026), sole author of **four manuscripts** and co-founder of Gigzen. Six fixes merged in 11 days into **Apple MLX**, **Meta ExecuTorch**, **Hugging Face OpenEnv**, **Oxford's PyZX** and **JaxMARL**. I build systems from scratch and benchmark them against the leaders: GPU kernels at **cuBLAS** parity, an LLM engine matching **vLLM** token for token, a collision-risk engine matching **NASA CARA** on all 53 of its test cases, and a 494M-parameter LLM that gives **identical bits on an NVIDIA GPU and a CPU**.

OPEN-SOURCE IMPACT

Merged: six fixes in 11 days (16–26 Oct 2026), each reviewed by the project's own maintainers

- **Apple MLX** #4737 (Apple's ML framework for Apple silicon, **29k★**): fixed a data race in the distributed ring backend, the TCP transport that links Macs for multi-machine training: its socket threads read their task queues outside the lock while other threads appended to them. A ThreadSanitizer harness found 2–4 races on every run before the fix and **0** after; MLX's ring tests pass on 8 ranks. Reviewed, approved and merged by MLX maintainer @zcbenz, who also pushed a follow-up commit (22 Oct).
- **Meta ExecuTorch** #23677 (on-device AI in Instagram, WhatsApp, Facebook, Messenger, Quest and Ray-Ban Meta, **5.1k★**): the 2- and 4-bit quantized embedding kernels of on-device LLMs read memory outside the table on an invalid token id and returned it as an embedding; added the bounds check the 8-bit kernel has, with tests that fail without it (quantized suite: 82 of 82 pass). Approved by Meta maintainers Scott Roy ("LGTM! Thanks for the fix!") and Jacob Stevens; merged 16 Oct.
- **Oxford PyZX** #563 (the ZX-calculus quantum-circuit optimiser by Kissinger and van de Wetering, **553★**): found by checking optimised circuits against the originals that three optimisers put an X rotation on the wrong qubit after rewriting two CNOTs as a SWAP, changing the unitary of **25 of 100** random Clifford circuits (0 after the fix); all 5 automated checks passed. Merged by co-creator John van de Wetering, 22 Oct: "Thanks for the fix! This was a regression from an earlier PR."
- **JaxMARL** #206 (NeurIPS 2024 Datasets and Benchmarks, **868★**; BOLD Lab: Oxford, UCL & Imperial): the rollout manager behind 8 Q-learning baselines (QMIX, VDN, IQL, PQN-VDN, SHAQ, TransfQMIX) zeroed the first observation of every new episode after an auto-reset, so agents chose their first action blind and trained on it; fixed while keeping zeros for agents that die mid-episode, and two of three new tests fail without the fix. Approved and merged by lead author Alex Rutherford, 26 Oct: "Good find!!"
- **Hugging Face OpenEnv** #1441, #1442 (RL post-training environments governed by Meta-PyTorch, Hugging Face, NVIDIA, Microsoft and others, **2.8k★**): fixed composite reward rubrics for RL training that ran pre-hooks after scoring their children (a timer read 0.01 ms for 400 ms of work) and scored an async judge as a coroutine; 9 of 13 new tests fail without the fixes. Both merged 19 Oct: "177 rubric tests pass".

In review

- **Google JAX** #41859 (**36.5k★**; Google assigned it to JAX co-author Matthew Johnson, 21 Oct): jax.grad through lax.reduce_window with a custom reduction has failed since 2021; its derivative is now a fold over the window offsets built only from transposable primitives, in the order XLA evaluates the reduction; 2,097 cases pass in 32-bit and 2,578 in 64-bit, and the 242 new ones fail on main.
- **Meta ExecuTorch** #23912 (revised and rebased after Meta maintainer review): zero points were checked against the output type, not the scales' type, so valid fp16-scale models aborted, and the .out variants read past their scales (AddressSanitizer: heap-buffer-overflow); the 2- and 4-bit export kernels now also report the scales' dtype; 86 of 86 kernel tests pass, and the new ones fail without the fixes.
- **vLLM** #60955 (**95.7k★**; labelled a bug for its NVIDIA project): FlashInfer prefill tests compared non-causal attention with a causal reference; against a float64 reference, all 32 cases now pass and all 32 fail on the bug.
- **NVIDIA Ising-Decoding** #98: the public-model tests, smoke test and docs looked only for .pt checkpoints, so anyone using the SafeTensors models NVIDIA publishes on Hugging Face failed them; they now accept the published files.
- **PyTorch** #199875, #199441, #199646: torch.compile now fills torch.empty in deterministic mode as eager does, without filling Inductor's own buffers (why an earlier fix was reverted); checked all 37 Inductor-skipped tests on CPU and two T4s, removing 13 stale skips and re-enabling 8 determinism tests, with verified causes for the 11 that stay skipped.
- **MIT Han Lab** nunchaku #957: a lone 4-bit LoRA ignored strengths below 1. **Carnegie Mellon** XGrammar #1012: the structured-output engine in vLLM and SGLang crashed when a grammar forced part of a UTF-8 character.
- **ETH Zurich** DaCe #2729: compiled numpy.rint rounded halves away from zero (2.5 → 3; NumPy gives 2); also filed a NaN bug in max/min (#2730).

Reported upstream

- **IBM Relay-BP** #47: reported that IBM's reference decoder gives different answers for the same shot depending on batch order, with a reproduction script and two candidate fixes. **Riverlane Deltakit** #411: proposed offline Relay-BP decoding for its qLDPC circuits.

RECOGNITION

- **NASA CARA** (Conjunction Assessment Risk Analysis): its operations team lead referred my independent reproduction of CARA's Monte Carlo results on all 53 of its test cases to CARA's analysis team, which will report its findings (20 Oct 2026; emails).
- **Entrepreneur First** (UK): its Talent Investor scouted me from public work alone (15 Oct 2026), citing my cuBLAS-parity compiler and a negative result I chose to publish, "which I rarely see" (email).

MANUSCRIPTS AND RELEASES

- S. Kundu. "Removing the Token Tax from an Existing Model: A Vocabulary Extension for 24 Indian Languages That Keeps Every Other Token." Manuscript, 2026; arXiv (cs.CL) submission in progress (paper).
- S. Kundu. "The Sampler Was Not the Policy: Finding and Removing a Silent Training–Inference Mismatch in a From-Scratch GRPO Stack." Technical report, 2026; arXiv (cs.LG) submission in progress (paper).
- S. Kundu. "Reproducing NASA CARA's Monte Carlo Probability of Collision: An Open Two-Body Implementation, Checked on All 53 of CARA's Test Conjunctions." Manuscript, 2026; arXiv (physics.space-ph) submission in progress (paper).
- S. Kundu. "Scam Defence Stops at English: Measuring the Multilingual Protection Gap and Linking Scam Campaigns Across Languages." Manuscript in preparation, 2026.
- Released lipi-Indic 96K (on Hugging Face) and samebit, both Apache-2.0.

ML SYSTEMS: TRAINING, INFERENCE AND RL

ratchet — PyTorch, C++/CUDA: reinforcement-learning post-training (GRPO) from scratch · study page

- Lifted Qwen2.5-0.5B on GSM8K from **48.1% to 52.5%** in 100 steps on 2× T4, scoring all 1,319 test problems, with rollouts on relay and training on tandem.
- Cut weight sync **369×** with in-place GPU updates, and made steps **1.9×** faster with partial rollouts and one-step-ahead training.
- Found fp16 weight sync silently erasing most updates, so the sampler ran a policy the trainer never had (gap up to **5.2 nats**); mixed-precision rounding cut it to **0.04**.

relay — C++, CUDA, Swift: a disaggregated, fault-tolerant LLM inference cluster · study page

- Wrote flash-attention and flash-decoding kernels that cut 2,048-token prefill **8.7×** (3.7 s → 0.43 s) and batch-32 decode **5.4×** on a T4; output matches Hugging Face.
- Streamed the KV cache layer by layer on a separate copy stream, hiding **99–100%** of the transfer: a 46 MB hand-off went from 21.6 to **0.08 ms**.
- Kept output identical through failures: watchdogs reroute requests from killed, frozen or stalled workers (**384/384 through 60 kills**; none lost in 300 fault-injection seeds).

samebit — C, CUDA, PyTorch, MLX: LLM inference in integers, the same bits on every chip

- Ran **Qwen2.5-0.5B** (494M parameters, 24 layers, grouped-query attention) **entirely in integers** (int8 weights, int16 activations, int32 residual stream, int64 sums): WikiText-2 perplexity **23.16 vs 22.99** for float32 (**+0.75%**); 93.25% of top-1 tokens unchanged.
- Showed **bit-identity on an NVIDIA Tesla T4**: it reproduces the x86 CPU's logit digests on real text, with each dot product split over 1–64 blocks and summed by atomicAdd in a new order every run; float16 shifts logits by up to **0.25**.
- Made **six independent backends** (C99/OpenMP, NumPy, PyTorch, **Apple MLX**, **CUDA**, reference) give **one digest over all 153 intermediate tensors** on x86-64, ARM64 and the T4, checked in CI on every push.
- Proved no value overflows in any summation order (exact integer-sqrt RMSNorm; tables built with 60-digit decimals). Cost: **531 tokens/s** on the T4 vs 6,033 for float16; int8 tensor-core kernels are next.

switchyard — CUDA, C++: mixture-of-experts serving with expert parallelism · study page

- Wrote routing, sorting and grouped expert GEMM/GEMV kernels on tensor cores and peer-to-peer dispatch: OLMoE-1B-7B decodes **2.2× faster than Hugging Face**, bit-identical to one GPU, with dispatch **1.25× faster than NCCL**.
- Showed on real text that dropping tokens at full experts raises perplexity from 12.9 to 20.2 even at capacity factor 2, so the kernels never drop one; found and fixed two cross-GPU stream races.

crucible — Python on Linux: sandboxed environments for RL on coding agents · study page

- Built sandboxes from namespaces, seccomp-BPF and limits that fork in **4.5 ms** (vs 138 ms to rebuild); its grader passes **0 of 9** known reward hacks (usual: 4).
- Trained with GRPO on ratchet, every reward from a sandbox: MBPP pass@1 **34.9% → 38.6%** (3 seeds; paired test, $p = 0.013$).

tandem — C++, CUDA: distributed training without torch.distributed or NCCL · study page

- Built DDP, ZeRO-1/2/3, GPipe and 1F1B on a custom ring all-reduce that trains **bit-identically to PyTorch DDP/FSDP on NCCL**; ZeRO-3 cuts per-GPU memory 17% on an 86M-parameter GPT.
- Reached **within 2% of NCCL's** all-reduce bandwidth at 64–256 MB with lock-free rings, CUDA IPC and a fused reduce kernel.

GPU KERNELS AND COMPILERS

tessel — Rust, zero dependencies: a tile language and compiler for GPU and TPU kernels · study page

- Wrote a Triton-style compiler: typed tile IR, linear layouts, tensor-core CUDA (ldmatrix, mma.sync, swizzled shared memory, software pipelining), an auto-tuner, and Metal and TPU (Pallas) backends.
- Matched **cuBLAS** on GEMM (**1.00–1.04×**) and beat PyTorch SDPA on causal attention (**1.3–1.9×**), torch.compile on RMSNorm (1.27×) and Triton on softmax (1.07×) on a T4, checked against fp64.
- Built an LLM engine on these kernels alone (paged KV cache, continuous batching, CUDA graphs) that matches **vLLM token for token at 1.05× its throughput** on TinyLlama-1.1B.

kiln — Rust, zero dependencies: an ML compiler from ONNX to SIMD C and CUDA · study page

- Compiles ONNX graphs to fused SIMD C and CUDA (NVRTC) with matmuls auto-tuned on the target machine; output matches PyTorch.
- Gave the lowest latency on all three benchmark models on 4 CPU cores: BERT in **117 ms** vs 149 for torch.compile, 153 for ONNX Runtime and 189 for JAX/XLA; on a T4, SmolLM2 **1.65×** faster than torch.compile.

SPACE SAFETY AND QUANTUM ERROR CORRECTION

kessler — C++/CUDA, Python: an independent auditor of space traffic · live globe

- Matched **NASA CARA's** Monte Carlo collision probability on **all 53** of its test conjunctions (χ^2 $p = 0.67$), including the 28 where the standard 2D method is wrong, **15–19× faster** per core than NASA's library.

- Screens all **531 million pairs** of 32,593 tracked objects a day ahead in **28 s** on a T4, matching CelesTrak SOCRATES to 0.24 m (median); SGP4 runs 125.7M orbit states/s in fp64.
- Publishes a daily 3D globe of every active satellite's close approaches from a GitHub Actions pipeline, with a maneuver detector that uses debris as its control group.

syndrome — C++, CUDA: an open GPU decoder for IBM's quantum LDPC codes · study page

- Matched IBM's Rust Relay-BP reference **bit for bit on 256/256 shots** of the [[144,12,12]] gross code (391,320 edges), every float64 and float32 posterior included.
- Decoded a single shot **28.7× faster than IBM's decoder** (median 37.5 ms vs 1.08 s) and a batch 4.0× faster than IBM's on all CPU threads, after reproducing IBM's published error rate (41 vs 44 in 10,000).

parity — Python, Stim, PyTorch: an independent benchmark of quantum error-correction decoders

- Ran NVIDIA's two published Ising pre-decoders through NVIDIA's own pipeline (d = 5–13, 524,288 shots each), matching its logged error rates to every printed digit, against PyMatching on identical samples.
- Reproduced PyMatching 2 / Sparse Blossom (Higgott & Gidney): scaling exponent 1.38 (paper: 1.32), thresholds at 0.61–0.80% for d = 3–11, and 0 logical errors in 1M shots per basis at d = 17.

LANGUAGE AI

lipi — PyTorch: cutting the token tax on Indian languages · study page

- Measured 24 Indian languages on 15 tokenizers (Odia: 4.99× English on GPT-4o) and built lipi-Indic (on Hugging Face), which needs fewer tokens than **GPT-4o's and Gemini's in 23 of 24** (1.30× vs 1.92×, 1.58×).
- Kept English, code and other scripts on exactly their Qwen 2.5 token ids, since the 96K new merges fire only on Indian-script bytes; Santali drops from 13.7× English to 1.52×.
- Extended Qwen2.5-0.5B and continued pre-training on 2× T4: Odia takes **8.6×** fewer tokens and, in equal GPU time, is predicted **20% better**, with Odia→English chrF++ 16.0 → 24.8.

babelguard — Python: an open early-warning system for scams that cross languages (in progress) · study page

- Showed in 33,869 scam SMS across 49 languages that an English keyword filter catches **86%** of English scams but **10–36%** of Korean, Japanese, Urdu, Arabic, Thai and Hindi ones.
- Built linking by meaning that recognises a translated scam **85%** of the time (text matching: 1.5%), on a crash-safe streaming engine: **4,057 messages/s** on 4 cores, nothing lost in 50 kills.

DISTRIBUTED SYSTEMS AND VERIFICATION

quorumdb — Rust, zero dependencies: a distributed SQL database, PostgreSQL wire-compatible · study page

- Built LSM storage, self-rebalancing multi-Raft, serializable ACID transactions (Percolator) and a SQL planner that psql connects to; **1,500 tpmC** on a TPC-C-derived benchmark.
- Found **6 real bugs** with deterministic simulation (**10,000 Raft clusters, 610K crashes**, histories machine-checked) and a TLA+ model (337K states), which also catch **all 15 planted bugs**.

faultline — Rust, zero dependencies: Jepsen-style fault injection for etcd

- Checked 1.26M operations under 288 faults across 4 etcd release lines for linearizability, and traced lost updates under etcd locks to a paused leader applying a timed-out write 6 s late; fenced writes lost none.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer **Apr 2026 – Present**

Early-stage startup (company name approved by India's MCA; incorporation in progress) · Sole engineer across every product

- Built **Waggle**, a delivery platform of three Android apps (riders, shops, customers) from one TypeScript/React/Capacitor codebase on Supabase PostgreSQL: **53 tables, 90 RLS policies, 154 SQL functions**.
- Moved first-accept-wins dispatch, UPI reconciliation and GST billing into the database, where no client can bypass them, and tested its access rules with **150 automated attacks**, all blocked.
- Released **Populace**, an open-source tool that fills an app with simulated users acting as real authenticated accounts through its own API (npm, **1,600+ downloads**, zero runtime dependencies, 122 self-tests); its first run found **5 defects** in 3.5 minutes after manual testing had passed, including a permission rule that silently stopped every new signup from creating a profile.
- Scaled it to **932,455 API calls** from 200 simulated drivers across 20 cities in 11 countries with no failures, and **3.17 million calls in a day** locally; pointed at Gitea's 482-operation API, it ran 1,106 calls cleanly and found 5 defects in its own adapter generator.
- Shipped all three Waggle apps signed for Android 7.0+ (Waggle Gig 1.6.2, Waggle Business 1.2.0, Waggle 1.1.0), now in closed testing, with 42 steps clicked through all three live from rider verification to the final rating.

Freelance Data Scientist — Upwork, remote (client in Spain) **Jun 2026 – Jul 2026**

- Merged two hotel censuses (5,909 and 8,058 records) and built a Google Trends panel of 10,038 city-months in Pandas.

SKILLS

GPU and compilers: CUDA (tensor cores, mma.sync, ldmatrix, graphs, IPC), NVRTC, Triton-style tile languages, Metal, Pallas, SIMD
ML systems: GRPO, DDP, ZeRO-1/2/3, FSDP, 1F1B, MoE expert parallelism, paged KV cache, int8 quantization, deterministic inference
Distributed systems: Raft, multi-Raft, MVCC, Percolator, LSM trees, deterministic simulation, TLA+, fault injection
Languages and platforms: C++20, CUDA, C, Rust, Python, Swift, TypeScript, SQL; PyTorch, JAX, MLX, PostgreSQL, Docker, Kubernetes

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata **Sep 2022 – Apr 2026**